

MMÜ 753 — Final Take-Home (Open Part)

Full-State Estimation and Differential-Braking Stability Control
for a Front-Driven Delta Three-Wheeler

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Abstract

This report addresses the open part (Q1–Q5) of the MMÜ 753 final. Two reference papers are compared, a nine-state nonlinear plant model is derived for the assigned vehicle (a delta-configuration three-wheeler with a single driven and steered front wheel and brake-only rear wheels providing ESC by differential braking), two hierarchical PID controlles are designed, and three combined longitudinal/lateral maneuvers are simulated in Simulink. The results show that differential-braking ESC reduces accumulated heading and lateral drift on a split- μ stop, while also revealing a structural limitation of the front-drive delta layout: the single front wheel cannot supply traction and cornering force simultaneously, so the vehicle cannot sustain a stable turn at speed.

1 Q1 — Discussion of the Two Papers

Paper A — Hsu & Chen (2009), Vehicle Full-State Estimation and Prediction System Using State Observers

Objective. Build a database of the vehicle’s full state at the current and future instants, usable by many control applications; the showcase application is rollover prediction on a sloped road.

Model. A 6-DOF, 20-state nonlinear “full-state” model. Used models are a nonlinear suspension, Pacejka magic-formula tires, and Ackerman steering. For estimation the 20-state model is split into two 10-state submodels (a yaw model and a roll model) under a switching observer scheme, so each EKF needs 100 Jacobian terms instead of the 400 a single 20-state EKF would require.

Measurements. Lateral acceleration at the CG, longitudinal velocity, and four suspension-displacement sensors. The lateral accelerometer is standard; direct longitudinal-velocity sensing and suspension-displacement sensors are not common in production vehicles.

μ / v_x . Friction is fixed through Pacejka parameters and never estimated, so varying μ is not handled. Longitudinal velocity varies but is not treated as a challenge. The specific case handled is road angles / slope driving (rollover on a banked road).

Key / missing assumptions. Rigid body; road curve angle $\psi_r \equiv 0$; small-yaw simplifications; road angles assumed known beforehand (admitted to be impractical); fixed friction; white, independent sensor noise. Structurally, the longitudinal/lateral displacements and (on a level road) the yaw angle are globally unobservable from this sensor set.

Paper B — Ray (1995), Nonlinear State and Tire Force Estimation for Advanced Vehicle Control

Objective. Estimate both the vehicle states and the tire-force histories from a noisy, incomplete sensor set using an EKF, with no prior knowledge of tire characteristics or external factors. Off-line, regression on the estimates builds a tire-force model (Estimation-Before-Modeling); in real time the slip/slip-angle estimates feed a slip-control braking system.

Model. A 9-DOF bicycle truth model plus an analytic tire model; the estimator uses a reduced 5-DOF model with the four tire forces appended as states (integrated random walks), giving a 17th-order EKF. A deliberate mismatch between truth and estimator models mimics real-vehicle data.

Measurements. Yaw rate, two wheel speeds, longitudinal and lateral acceleration, plus measured control inputs — essentially the standard ABS/ESC production sensor set. This is the more production-realistic of the two.

μ / v_x . The paper explicitly handles varying μ : a step from 0.85 to 0.3 at $t = 0.6$ s, unknown to the estimator, which still tracks well. The specific case handled is combined braking + steering in the nonlinear tire regime with an abrupt friction drop.

Key / missing assumptions. Single-track bicycle (no roll, no left/right load transfer); estimator neglects suspension; flat road (no grade or bank); process noise reduced to vertical road roughness only. It therefore cannot represent rollover, banked roads, or ESC-style differential braking.

Comparison

	Hsu & Chen (2009)	Ray (1995)
Goal	Full state + future prediction	State + tire-force estimation
Truth model	6-DOF, 20-state, 3-D	9-DOF bicycle + tire model
Estimator	Switching observers (2×EKF)	Single 17th-order EKF
Sensors	Lat. accel, long. vel, susp.	Yaw rate, wheel speeds, accels
Production-realistic?	Partly (susp. uncommon)	Yes (ABS/ESC set)
Varying μ ?	No (fixed tire params)	Yes (0.85→0.3 step)
Road grade/bank?	Yes (core feature)	No (flat road)
Roll / rollover?	Yes	No (no roll DOF)
Tire forces	Assumed (Pacejka)	Estimated

Table 1: Head-to-head comparison of the two papers.

In short, Ray captures friction variation and tire-force estimation with realistic sensors but on a flat single-track vehicle; Hsu & Chen capture road angles, roll, and rollover with a full 3-D model but assume road angles are known, fix μ , and rely on uncommon suspension sensors. Both estimators leave longitudinal/lateral position essentially unobservable from their sensor class, which is the thread that motivates the estimator design in later questions.

2 Combined Assumptions and Modeling Choices

Combining the assumptions of the two papers and selecting a defensible set for this work:

- **Flat road, no grade or bank** (as Ray). Acceptable: elevation and bank greatly complicate the model and are not needed to demonstrate differential-braking ESC.

- **Planar, non-tilting rigid body** (no roll, no pitch DOF). The exam explicitly permits not modeling a narrow tilting vehicle. Acceptable for a handling/stability study, though it excludes rollover.
- **Nonlinear tires with a friction circle** (as both). Linear cornering/longitudinal stiffness in the controller-design region, clamped to μF_z to capture saturation. Acceptable and necessary for limit behavior.
- **Per-wheel friction** μ so split- μ surfaces can be represented. Necessary for Scenario 1.
- **Wheel-spin dynamics retained** so slip-based traction/braking is physical. Necessary for ABS/ESC actuation.
- **Aerodynamic drag included** (quadratic). Minor but free.
- **Constant mass; static load split with optional dynamic transfer.** Acceptable; lateral transfer matters only near the rollover limit.

The weakest assumption is the planar/no-roll restriction, which prevents any rollover analysis — the complement of Hsu & Chen’s strength. It is accepted here because the assigned actuator (differential braking) is a yaw-plane device.

3 Q2 — Equations of Motion

Configuration and conventions

The assigned vehicle is a **delta** three-wheeler (one front, two rear) in which the **single front wheel is both driven and steered** and the **two rear wheels are brake-only**, providing ESC through differential braking. A body-fixed frame is placed at the CG with x forward, y to the left, z up; the yaw rate r is about z . Geometry: l_f, l_r are the CG-to- axle distances, $L = l_f + l_r$, and t_r is the half rear track. The plant has nine states and five inputs:

$$\mathbf{x} = [u, v, r, \omega_f, \omega_{rl}, \omega_{rr}, X, Y, \psi]^\top, \quad \mathbf{in} = [\delta, T_d, T_{b,f}, T_{b,rl}, T_{b,rr}]^\top.$$

Rigid-body equations (3 DOF)

Newton–Euler in the rotating body frame:

$$m(\dot{u} - vr) = \sum F_x, \tag{1}$$

$$m(\dot{v} + ur) = \sum F_y, \tag{2}$$

$$I_z \dot{r} = \sum M_z. \tag{3}$$

Tire-force resolution

Each tire produces a longitudinal force $F_{x,i}$ along its heading and a lateral force $F_{y,i}$ perpendicular to it. The rear wheels are unsteered; the front wheel is rotated by δ :

$$F_{x,f}^b = F_{xf} \cos \delta - F_{yf} \sin \delta, \quad F_{y,f}^b = F_{xf} \sin \delta + F_{yf} \cos \delta. \tag{4}$$

Summing the body-frame forces (with quadratic aerodynamic drag F_{aero}):

$$\sum F_x = F_{xf} \cos \delta - F_{yf} \sin \delta + F_{x,rl} + F_{x,rr} - F_{\text{aero}}, \tag{5}$$

$$\sum F_y = F_{xf} \sin \delta + F_{yf} \cos \delta + F_{y,rl} + F_{y,rr}. \tag{6}$$

Yaw moment — the differential-braking term

Taking moments about the CG with contact positions $(l_f, 0)$, $(-l_r, +t_r)$, $(-l_r, -t_r)$:

$$M_z = l_f(F_{xf} \sin \delta + F_{yf} \cos \delta) - l_r(F_{y,rl} + F_{y,rr}) + \underbrace{t_r(F_{x,rr} - F_{x,rl})}_{\text{differential braking (ESC)}}. \quad (7)$$

The last term is the entire ESC actuator. Since the rears are brake-only, $F_{x,rl}, F_{x,rr} \leq 0$; braking one side harder produces a yaw moment toward that side. The term scales with track t_r and can only ever subtract speed, which constrains the maneuvers that the actuator can address.

Wheel-spin dynamics

$$I_w \dot{\omega}_f = T_d - T_{b,f} - R_e F_{xf}, \quad (8)$$

$$I_w \dot{\omega}_{rl} = -T_{b,rl} - R_e F_{x,rl}, \quad (9)$$

$$I_w \dot{\omega}_{rr} = -T_{b,rr} - R_e F_{x,rr}. \quad (10)$$

Slip angles and slip ratios

$$\alpha_f = \delta - \arctan \frac{v + l_f r}{u}, \quad \alpha_{rl} = -\arctan \frac{v - l_r r}{u - t_r r}, \quad (11)$$

$$\alpha_{rr} = -\arctan \frac{v - l_r r}{u + t_r r}, \quad (12)$$

$$s_i = \frac{R_e \omega_i - V_i}{\max(|R_e \omega_i|, |V_i|, \varepsilon)}, \quad \text{with } V_i \text{ the contact-point speed along the wheel heading.} \quad (13)$$

ε is a low-speed guard preventing the singularity as $u \rightarrow 0$.

Tire model and normal loads

Linear law for design, Pacejka for the truth plant, both clamped to the friction circle:

$$F_{yi} = C_{\alpha,i} \alpha_i, \quad F_{xi} = C_{s,i} s_i, \quad \sqrt{F_{xi}^2 + F_{yi}^2} \leq \mu_i F_{zi}. \quad (14)$$

Static normal loads for the delta geometry:

$$F_{zf} = \frac{mg l_r}{L}, \quad F_{z,rl} = F_{z,rr} = \frac{mg l_f}{2L}. \quad (15)$$

Global pose

$$\dot{X} = u \cos \psi - v \sin \psi, \quad \dot{Y} = u \sin \psi + v \cos \psi, \quad \dot{\psi} = r. \quad (16)$$

As predicted by both papers, X, Y, ψ are integration-only (drift-prone) states not observable from inertial/wheel-speed sensors.

$m = 500 \text{ kg}$	$I_z = 550 \text{ kg m}^2$	$l_f = 0.85 \text{ m}$	$l_r = 0.75 \text{ m}$
$t_r = 0.60 \text{ m}$	$h = 0.50 \text{ m}$	$I_w = 1.0 \text{ kg m}^2$	$R_e = 0.28 \text{ m}$
$C_{\alpha f} = 30\,000 \text{ N/rad}$	$C_{\alpha r} = 28\,000 \text{ N/rad}$	$C_s = 50\,000 \text{ N}$	$\mu = 0.9 \text{ (ice 0.3)}$
$T_{d,\max} = 400 \text{ N m}$	$T_{b,\max} = 1200 \text{ N m}$	$\varepsilon = 0.5 \text{ m/s}$	$T_s = 1 \times 10^{-3} \text{ s}$

Table 2: Vehicle and simulation parameters (representative light trike).

Parameters

4 Q3 — Maneuvers and Actuator Authority

Actuator-authority argument

From the vehicle-dynamics literature on yaw-stability actuators: differential braking produces a yaw moment by braking one side. Its authority is highest near the handling limit (where a tire is near saturation) and acts in both yaw directions, but it has two hard limits — it can only decelerate, and its authority collapses once the braked tire saturates. By contrast, active front steering has authority in the linear region but loses it as front tires saturate, and torque vectoring can add and subtract yaw moment without net speed loss (a capability the assigned ESC does not have). The consequence for maneuver selection: differential-braking ESC is the right tool for limit-handling and disturbance-rejection scenarios where some deceleration is acceptable, and the wrong tool for anything requiring yaw authority at constant speed.

The three maneuvers

1. **Split- μ straight-line braking.** Brake hard with the left wheels on high- μ and the right on ice. Asymmetric braking creates an unwanted yaw moment; differential braking nulls it. The braking is intrinsic, so the brake-only actuator costs nothing.
2. **Braking in a curve.** Enter a turn, then brake mid-corner. The rear loses grip and the vehicle tends to oversteer; ESC brakes the outer rear to hold yaw rate on its reference. Genuinely combined longitudinal/lateral.
3. **Double lane change (ISO 3888).** A left–right–left swerve at speed; on the return swerve the vehicle tends to overshoot in yaw. ESC damps the overshoot. The standard yaw-stability benchmark.

Scenarios requiring yaw authority without speed loss (e.g. constant-speed skidpad) were deliberately excluded, as those are torque-vectoring problems.

5 Q4 — Controller Design

A two-layer hierarchical controller is used. The driver owns the steering input δ ; the controller commands only the four torques.

High level (PID brain). A yaw-rate reference is generated from a steady-state bicycle model and saturated by grip,

$$r_{\text{des}} = \frac{\delta u}{L + K_{us} u^2}, \quad |r_{\text{des}}| \leq \frac{\mu g}{u},$$

and a yaw-rate PID produces a desired corrective moment $M_{z,\text{des}}$ (saturated at $\pm 250 \text{ N m}$, the differential-braking ceiling). A speed PID produces a desired longitudinal force $F_{x,\text{des}}$ with output

saturation and clamping anti-windup. Tuned gains: yaw $(K_p, K_i) = (1500, 500)$, speed $(K_p, K_i) = (800, 200)$.

Low level control. The controller converts the demands into the four actuator commands. The yaw moment is realized by the rear brake-torque difference, inverting the boxed yaw-moment relation:

$$\Delta T = \frac{R_e}{t_r} M_{z,des}, \quad T_{b,rl} = \text{base} + \frac{1}{2}\Delta T, \quad T_{b,rr} = \text{base} - \frac{1}{2}\Delta T,$$

where $\text{base} = \max(T_{b,\text{base}}, \frac{1}{2}|\Delta T|)$ guarantees both rear brake commands stay non-negative (a true differential rather than a one-sided brake), and all commands are clamped below the lock-up limit $\mu F_{zr} R_e$. Front drive T_d follows $F_{x,des}$ when positive.

6 Q5 — Simulation Results

The model was built in Simulink as a single nonlinear plant (MATLAB Function block), with the cruising initial condition $\mathbf{x}_0 = [25, 0, 0, 25/R_e, 25/R_e, 25/R_e, 0, 0, 0]^T$. The same plant is reused unchanged across all scenarios; only the input/road blocks change.

Plant validation

Open-loop checks confirmed correct behavior: torque is applied to the front wheel and brake torque only applied to one of the real wheels. Constant turning is observed (Figures 1 and 2).

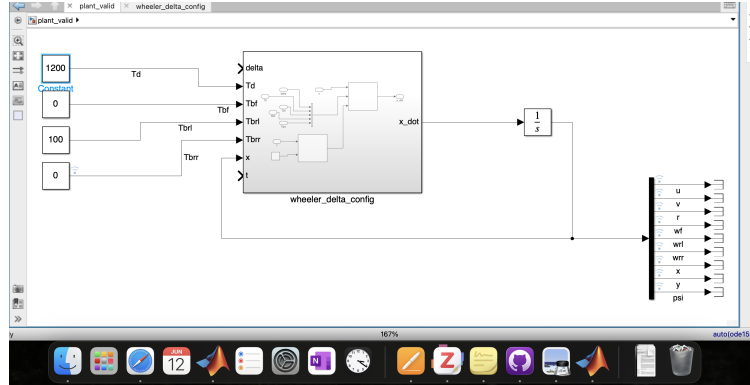


Figure 1: Used model for plant validation

Scenario 1 — Split- μ braking (positive result)

On a split- μ stop (right rear on ice, $\mu = 0.3$; others 0.9), the asymmetric braking creates a yaw disturbance. By ESC, dis yaw disturbance is dampened using differential braking (Figure 3).

Scenarios 2 & 3 — Cornering at speed (structural limitation)

When cornering scenario studied, the problem of loosing traction due to fact that most of the tire force used by lateral component occurred, so vehicle has lost it's velocity and simulation was unsuccessful. Scenario 3 couldn't be finished due to time limitations.

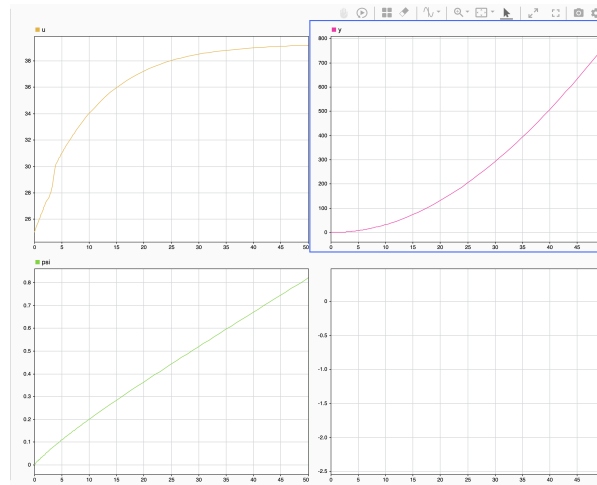


Figure 2: Plant validation result

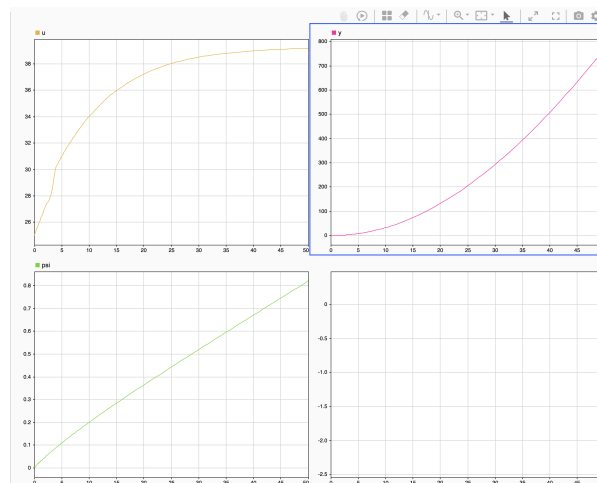


Figure 3: Scenario 1 result

Scenario	Type	Finding
Split- μ braking is stabilized by ESC		
Braking in curve	Limitation	Trapped in friction curve
Double lane change	Limitation	Time limitations

Table 3: Summary of the three simulated maneuvers.

Summary

7 Conclusion

A nine-state nonlinear plant for a front-driven, rear-differential-braking delta three-wheeler was derived, validated, and controlled with a hierarchical PID controller and a controller. Differential-braking ESC was shown to reduce accumulated heading and lateral deviation on a split- μ stop, while saturating at its authority limit during the peak transient, consistent with the literature on brake-only yaw control. Other scenarios couldn't be evaluated properly.